

AYUSH CHAUHAN

Research Portfolio | Civil Engineering Punjab Engineering College (Deemed to be University), Chandigarh-160012, India **Email:** chauhanayush485@gmail.com | **ORCID:** [0009-0000-4189-7676](https://orcid.org/0009-0000-4189-7676)

1. RESEARCH SUMMARY

Final-year B.Tech Civil Engineering researcher specializing in **Sustainable Urban Systems, AI-driven Transport Planning, Environmental Data Modeling, and Spatial Analytics**. My academic work is deeply aligned with the United Nations Sustainable Development Goals (SDGs-2030), focusing on actionable, data-driven interventions for traffic management, pedestrian safety, air quality control, and ecological restoration in developing urban landscapes like Chandigarh and the Indo-Gangetic Plain.

Working collaboratively with Dr. Badveeti Adinarayana and other researchers at PEC and CSIR-CRRI, I have contributed to numerous peer-reviewed journals and international conferences, bridging the gap between traditional civil engineering and computational intelligence.

2. DETAILED RESEARCH PUBLICATIONS & MANUSCRIPTS

A. Environmental Modeling & Air Quality

Paper 1: Modeling of Seasonal Dynamics for Traffic-Dominant Particulate Matter Pollution in The Indo-Gangetic Plain: Using Multi-Scale Wavelet Analysis, Source Apportionment, And Episode-Specific Control Strategies

- **Journal/Target:** Sustainable Intelligent Infrastructure (SII)
- **Authors:** Dr. Adinarayana Badveeti, Ayush
- **Research Context:** The Indo-Gangetic Plain experiences drastic PM_{2.5} fluctuations, ranging from 140–180 $\mu\text{g}/\text{m}^3$ in winter to 35–55 $\mu\text{g}/\text{m}^3$ during the monsoon. Despite this 4.5–5x variance, air pollution regulations rarely adapt seasonally.
- **Methodology:** Investigated Delhi's air quality over a five-year period (2019–2024) utilizing Rolling Positive Matrix Factorisation (PMF), k-means clustering, continuous multi-scale wavelet analysis, and season-specific machine learning.
- **Key Findings:** * Wavelet analysis identified three primary rhythms: a seasonal cycle (~200 days), a transitional cycle (75–100 days), and a weather-driven cycle (10–40 days).
 - Rolling PMF revealed that traffic's contribution to PM_{2.5} surges to 35–50% during winter due to atmospheric stagnation, while dropping to 10–15% in the monsoon.
 - 70–85% of vehicle-induced PM originates from non-exhaust sources like road dust, tire wear, and brake dust.
 - Generic machine learning models ($R^2=0.74$) were outperformed by season-specific models (Winter $R^2=0.89$, Monsoon $R^2=0.93$).
- **Policy Impact:** Proposed targeted winter interventions (limiting heavy trucks, regulating heating) capable of reducing PM by 30–40%. Combined with post-monsoon biomass burning reductions, annual PM can be reduced by 40–50%, aligning with WHO 2030 targets.

B. Ecological Restoration & Remote Sensing

Paper 2: Sustainable Water Transport and Lake Restoration: Integrating AI and Satellite Data for Sukhna Lake in Chandigarh Green City, India for Achieving to SDGs-2030

- **Target:** 41st IAHR World Congress Submission
- **Authors:** Badveeti Adinarayana, Ayush
- **Research Context:** Addressing the pressing challenges of eutrophication and ecological degradation in urban lakes, specifically Sukhna Lake in Chandigarh.
- **Methodology:** Combined AI-driven predictive modeling with eco-innovative solutions utilizing satellite remote sensing data from Landsat-7 ETM+ and ASTER, alongside in-situ measurements.
- **Key Findings:** The framework successfully predicts critical water quality parameters essential for forecasting eutrophication, including Secchi Disk Transparency (SDT), Chlorophyll-a, and Land Surface Temperature (LST).
- **Proposed Interventions:** The study introduces floating treatment wetlands and bioremediation as nature-based, scalable, adaptive solutions to mitigate nutrient inflow and improve water quality globally.

Paper 3: AI-Driven Predictive Modelling and Sustainable Interventions for Lake Restoration: A Case Study of Sukhna Lake Using Satellite Data

- **Target:** CISSC 2025 Submission
- **Authors:** Ayush, Dr. B. Adinarayana
- **Research Context:** Urban lakes are critical resources for biodiversity that face severe threats from human activity and climate change.
- **Methodology:** Leveraged satellite imagery and data-driven models to track the spatial distribution of ecological degradation.
- **Key Findings:** Established an AI model that serves as a proactive, scalable tool for urban lake management and decision-making, offering insights into sustainable ecosystem management practices globally.

C. Urban Mobility & Pedestrian Safety (SDG-11)

Paper 4: Development of AI model for gender differences in bicycle mobility of green urban areas for recreational and educational institutions of cycling behaviour, patterns and constraints

- **Journal:** Innovative Infrastructure Solutions (Published: <https://doi.org/10.1007/s41062-024-01709-w>)
- **Authors:** B. Adinarayana, Ayush, Balkaran Singh Sohal, Aryan, Reetik Kumar, Shubham Sharma
- **Research Context:** Investigating non-motorized transport (NMT) as a sustainable mode of travel in Chandigarh, specifically looking at how different demographics interact with bicycle infrastructure.
- **Methodology:** Collected traffic flow data at intersections in various zones, conducted widespread online and field surveys, and analyzed NMT regulations established by local governmental entities.
- **Key Findings:** Identified distinct gender-specific differences in cycling patterns, motivators, and constraints experienced by male and female cyclists. Revealed that male cyclists predominantly used bicycles for commuting short distances, while identifying unique barriers for female cyclists.
- **Policy Impact:** Provided actionable policy recommendations to enhance cycling infrastructure and promote equitable, sustainable transport.

Paper 5: Modeling Pedestrian Walking Behavior Parameters and Pedestrian Safety at Congested Urban Intersections in Educational and Recreational Zones for SDGs-2030

- **Target:** SPARC Sponsored International Conference (SAFEIND 2026)
- **Authors:** Dr. Adinarayana Badveeti, Ayush, Aryan, Reetik Kumar, Kasinayana Badveeti
- **Research Context:** In well-planned cities like Chandigarh, pedestrian safety at intersections is heavily influenced by surrounding dynamics.
- **Methodology:** Modeled pedestrian safety and walking behavior by analyzing population density, land-use patterns, public transport availability, and surrounding activity intensity.
- **Key Findings:** Demonstrated that improving pedestrian safety requires moving away from generic solutions and actively integrating context-specific regional factors into traffic planning, design, and management.

Paper 6: Study of Pedestrian and Slow-moving Traffic Control Behaviour at Congested Intersections for Urban Areas in Educational and Recreational Zones to Achieve SDGs-2030 Targets

- **Journal:** Innovative Infrastructure Solutions
- **Authors:** Dr. Badveeti Adinarayana, Ayush, Aryan, Reetik Kumar
- **Research Context:** Enhancing pedestrian safety by evaluating the unique characteristics of congested intersections in developing urban centers to support Sustainable Development Goal 11 (Sustainable Cities and Communities).
- **Methodology:** Employed an extensive observational analysis methodology and raw data collection at intersections located within congested zones. Examined traffic flow, pedestrian density, and traffic signals.
- **Key Findings:** Generated valuable insights into the intricacies of navigating diverse urban landscapes, aiding city planners in designing safer and more efficient traffic control interventions.

Paper 7: Developing of Walkability Index for University Green Campus for the Evaluation of Existing Sustainable Pedestrian Walkability Infrastructure at PEC Campus

- **Target:** TPMDC 2024 Submission
- **Authors:** Dr. Badveeti Adinarayana, Ayush Chauhan, Aryan, Balkaran Singh Sohal
- **Research Context:** Evaluating how encouraging a specific area is to walking based on infrastructure, safety, accessibility, and environmental quality.
- **Methodology:** Developed a Linear Walkability Index (LWI) using a structured Google Survey format deployed to campus stakeholders. Metrics evaluated included lighting, presence of shade, crosswalk sufficiency, sidewalk width, green cover, and stress-relief perception.
- **Key Findings:** Provided a quantifiable metric for university planners to evaluate and upgrade existing sustainable pedestrian infrastructure to meet SDG-2030 targets.

D. Spatial Analysis & Rural Infrastructure

Paper 8: Terrain-Sensitive Accessibility Modelling for Rural Settlements: A GIS-AHP Framework with Participatory Calibration

- **Institution:** Developed in CSIR-Central Road Research Institute (CRRI)
- **Authors:** Ayush Chauhan, Adinarayana Badveeti
- **Research Context:** Conventional proximity-based accessibility measures fail to adequately capture the unique challenges of rural mobility in hilly and mountainous regions.

- **Methodology:** Developed a comprehensive framework integrating Geographic Information Systems (GIS) with the Analytic Hierarchy Process (AHP). Applied to the West Garo Hills district in Meghalaya (1,176 habitations, 501,294 residents). Utilized Shuttle Radar Topography Mission (SRTM) high-resolution elevation data.
- **Key Findings:** Conducted a structured stakeholder survey (49 respondents) to derive consistency-validated priority weights for both service types and travel factors, successfully capturing the complex interplay of terrain constraints and service priorities.

3. DETAILED CONFERENCE PARTICIPATION & SUBMISSIONS

My research has been actively submitted to and presented at major international and national forums focused on civil engineering, infrastructure, and environmental sustainability:

- **Sustainable Intelligent Infrastructure (SII) & SPACE 2026**
 - *Focus:* Smart infrastructure and environmental dynamics.
 - *Contribution:* Submitted comprehensive research on multi-scale wavelet analysis and seasonal dynamics of particulate matter pollution in the Indo-Gangetic Plain.
- **41st IAHR World Congress (Singapore, June 2025)**
 - *Focus:* Innovative Water Engineering for Sustainable Development.
 - *Contribution:* Submitted extended abstract on sustainable water transport and AI-driven lake restoration utilizing satellite data.
- **SAFEIND 2026 - SPARC Sponsored International Conference**
 - *Focus:* Safe System Perspective for Indian Mixed Traffic Conditions.
 - *Contribution:* Authored research on modeling pedestrian walking behavior parameters and safety at congested urban intersections.
- **CISSC 2025 (International Conference on Challenges and Innovations for Sustainable Smart Cities)**
 - *Focus:* Smart city development and sustainable urban planning.
 - *Contribution:* Submitted predictive modeling research targeting eutrophication and sustainable interventions for urban lakes.
- **TPMDC 2024**
 - *Focus:* Transportation Planning and Traffic Engineering.
 - *Contribution:* Developed and presented a novel Linear Walkability Index (LWI) for university green campuses.
- **Additional Academic Forums:**
 - **AVCC Systems Workshop (2025):** Participated at CSIR-Central Road Research Institute (CRRI).
 - **YPC-CTRG 2025:** Engagement in transportation research topics.
 - **CIDC-ICC 2025:** Focus on construction and infrastructure development.
 - **UPES HSFEA 2023:** Early engagement in Health, Safety, and Environmental aspects of engineering.

4. TECHNICAL SKILLS & METHODOLOGIES

- **Data Science & AI:** Machine Learning (Predictive Modeling), K-Means Clustering, Continuous Multi-Scale Wavelet Analysis, Rolling Positive Matrix Factorisation (PMF).

- **Spatial & Environmental Analysis:** GIS-AHP Frameworks, Satellite Remote Sensing (Landsat-7 ETM+, ASTER, SRTM Digital Elevation), Deweathering Analysis, Land Surface Temperature (LST) mapping.
- **Transportation Engineering:** Linear Walkability Index (LWI) Development, Traffic Flow Analysis, Pedestrian Safety Modeling (SDG-11 Focus), Non-Motorized Transport (NMT) evaluation, Participatory Stakeholder Calibration.